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ORIGINAL ARTICLE

The role of CIO in digital transformation: an exploratory study

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Abstract
With a continuous stream of emerging technologies, organizations are increasingly seeking to digitally transform their business towards ever efficient, intelligent, and self-learning business models. As the recognized head of IT, the Chief Information Officer (CIO) is often tasked with the responsibility of successfully implementing Digital Transformation (DT) initiatives. While there are independent studies on CIO role, and on the impact of DT, the role of CIO is not sufficiently examined in the DT context. This study fills this gap by conducting multiple case study of six organizations engaged in DT. Following the role theory, the study emphasizes the CIO as an actor focused on digital strategy and empirically delineates the multidimensional task demands associated with DT regarding the CIO role under the new strategic realities of the digital age. We find that while traditional CIO roles still apply, the DT context adds more nuances in such roles. Moreover, some new roles (e.g., legacy systems integration, risk management and IT security) have become crucial that are not adequately captured in extant literature on CIO roles.

Keywords Digital transformation · Chief Information Officer · DT · CIO · Leadership

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1 Introduction

The pervasive emergence of digital technologies has hugely impacted the business sphere (Cozzolino et al. 2018; Nambisan et al. 2019; Hanelt et al. 2021; Kindermann et al. 2021). Organizations are increasingly embracing digital technologies to build ever efficient, intelligent, and self-learning business models (Verhoef et al. 2021). Companies like Airbnb, Amazon, Netflix, and Revolut have transformed completely on all-digital revenue models, disrupting the established players in respective industries. Numerous existing businesses have adopted digitalization and transformed their organizational structures and managerial roles in recent years (Haffke et al. 2016; Sebastian et al. 2020a, b; Verhoef et al. 2021; Ronteau et al. 2022).

DT is a complex and multidimensional phenomenon (Chawla et al. 2022; Verhoef et al. 2021). It is inextricably linked to the organizational change (Hansen et al. 2011; Hanelt et al. 2021; Zaki 2019) having ramifications across individual, community, organizational and societal levels (Nambisan et al. 2019). Therefore, effective leadership forms the foundation for the digital journey in organizations (McCarthy et al. 2021). As the recognized head of IT, the Chief Information Officer (CIO)[1] is often tasked with strategizing, executing, and sustaining DT in their organizations. Successful execution of DT requires incorporation of multiple perspectives, including strategy, technology, business processes, diversified stakeholders, organizational culture, and talent, to mention a few (Chawla et al. 2022). This multidimensionality of DT often puts CIOs to the test by offering tradeoffs such as scalability vs cost, accessibility vs security, cloud vs on-premise infrastructure deployment, and outsourcing vs in-house teams. Therefore, despite significant investments and efforts, DT initiatives sometimes fail to deliver (Davenport and Westerman 2018). This makes the phenomenon a strong prospect for further investigation on how CIOs can contribute to ensure the success of DT initiatives.

The extant research has emphasized the evolution and growth of the CIO functions over the years (Ross and Feeny 1999; Chun and Mooney 2009; Peppard et al. 2011; Davis and McDonagh 2015; Hütter and Riedl 2017; Hillebrand and Westner 2021). In recent years, studies have also been conducted on DT and its implications for the organizations (Berman 2012; Fitzgerald et al. 2014; Hanelt et al. 2021; Warner and Wäger 2019). A number of recent studies (Capitani 2018; Kunisch et al. 2020; Hess et al. 2020; McCarthy et al. 2021) encourage exploring individual characteristics of digital leaders, their strategic leadership capabilities and environmental aspects in context of organizational DT. While there exist a number of consulting industry-based report conducting surveys among CIOs, CTOs, and CEOs, one does not get sufficiently detailed insights on the DT phenomenon due to inadequate sampling and the nature of the survey instrument. However, existing literature still lacks

[1] Whilst recent literature notes the position of Chief Digital Officer (CDO), all our case organisations had a person with a formal designation of CIO responsible for leading DT. There was no CDO position in the selected case organisations. Nevertheless, in the context of DT, the discussions and findings are relevant to whomsoever is leading DT.

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clarity regarding the ground-level empirical aspects on the contributions of the CIOs in the contemporary phenomenon of DT. Considering these observations, there is a need for in-depth empirical research into the contributions of CIOs to unveil various factors that they address in leading digital transformation. This study intends to fill this research gap by exploring the following research question: *How do CIOs contribute to digital transformation?*

To explore the phenomenon as it exists in real-world setting, we adopt multiple case study approach (Yin 2017) by conducting semi-structured interviews with multiple stakeholders, including CIOs, business executives and technology partners of six large organizations. We consider this research to be significant in at least three aspects. First, our research provides detailed insights into the organizational elements that necessitate CIOs participation to organizational DT endeavors. Following the case analysis, the study identifies multidimensional task demands associated with DT. Hence, the study contributes to digital transformation research (Birkinshaw 2018; Lanzolla et al. 2018; Mayer 2021) by providing a glimpse into the micro-foundations of DT in large organizations. Second, the study contributes to research on the functional executive roles (Guadalupe et al. 2014; Firk et al. 2021; Menz 2021) specifically in context of CIO and DT, and may provide inspiration for future theory-testing investigations in this field. The research enriches the role theory literature by exploring and systematizing the CIO role interactions and actions through multiple disciplines to an integrated 'multi-role' understanding of the CIO role in the context of DT. We highlight the CIO as an actor focused on digital strategy by taking into consideration the new strategic realities of the digital age. In so doing, our study offers empirical understandings of the developing DT phenomenon with reference to the CIO role. Our exploratory investigation expands the insights offered by the relatively limited research in this field (Singh and Hess 2017; Tumbas et al. 2017; Capitani 2018; Singh et al. 2020; McCarthy et al. 2021). Third, our studies offers insights from a study conducted in an emerging economy. This contrasts with existing DT research that is largely conducted in advanced economies (Mićić 2017; El-Telbany et al. 2020; Firk et al. 2021; Kapucu 2021).

The conceptual foundation of this study is the role theory according to which, individuals act in ways that match the functional, relational, and structural characteristics of the social system with which they engage (Biddle 2013; Katz and Kahn 1978). Organizational personnel execute their work via fulfilling their roles (Graen 1976), and hence, role-play serves as a mechanism for establishing team-level procedures (Georgakakis et al. 2022). Role theory has three main schools of thought, functionalism, social interactionism, and structuralism (Biddle 1986). According to the functionalist perspective, an individual's predetermined functional titles establish their responsibilities in organized interactions (Bates and Harvey 1975; Luhmann 1995). From the social interactionism viewpoint, a social system's members' roles are iteratively constructed through continuous interactions, based on needs of individuals with in the organizations rather than position of individuals (Biddle 1986; Painter-Morland 2008; Raes et al. 2011). Finally, the structuralism viewpoint assumes that roles are defined by power relationships and disparities of actors in a social system (Bates and Harvey 1975; Biddle 1986; Dowding 2008). Role theory thus enables scholars in systematizing their assumptions about how individual

roles in communities are assumed and change over time (Biddle 2013). Due to its emphasis on social systems, role theory developments have become more prevalent in recent years to explain that the interactions among the actors of social system (organizations) occur to impact the organizational behavior and decisions (Biddle 2013; Georgakakis et al. 2022).

Consequently, there is a notable research strand that unveils leadership roles through the application of role theory. Many researchers and practitioners under information systems (Kunisch et al. 2020; Firk et al 2021; Kessel and Graf-Vlachy 2022; Lorenz and Buchwald 2023) as well as in management research have conceptualized executives' interactions and actions from a role theory perspectives in recent studies (Chang et al 2021; Holmes et al. 2021; Cortes and Herrmann 2021; Singh et al. 2023). CIOs and other executives [e.g., Chief Marketing officer (CMO), Chief Operating Officer (COO), Chief Human Resource Officer (CHRO)] have several, and frequently intersecting roles while performing their strategic organizational duties (Cannella and Holcomb 2005). Specifically in context of DT, this aspect outlined in role theory is expected to be relevant. As strategic leadership is a group effort (Simsek et al. 2018) epitomized by continuous interactions among the CIO and other actors (Bromiley and Rau 2016; Georgakakis et al. 2022), role theory can aid in systematically categorizing various assumptions for conceptualizing CIO role. Hence, we evaluate the CIOs activities through a theoretical lens of role theory to methodically outline different capacities through which CIOs practice strategic leadership for DT.

The remainder of this paper is organized as follows. A conceptual foundation of the DT and the literature on CIO role is presented in the literature review section. This is followed by details of the methodology used in this study. Thereafter, the derived CIO roles are presented in the findings section, followed by a comprehensive discussion in view of existing literature. Finally, we address research implications, limitations, and future scope.

2 Literature review

The term digital transformation (DT) represents the use of new-age digital technologies (for instance, big data, mobility, analytics, cloud computing, and internet of things) by the business to radically transform their customer offerings, operational processes, or the business models to generate new revenue streams (Berman 2012; Fitzgerald et al. 2014; Hanelt et al. 2021; Warner and Wäger 2019). DT also entails development of digital business models that allow for new methods based on data-driven insights to capture business value (Verhoef et al. 2021; Vial 2019; Ronteau et al. 2022). Hence, DT is not merely about technology; it is about the synergies that these new age digital technologies provide by combining information collection, processing, and communication via connected systems (Bharadwaj et al. 2013; Chawla and Goyal 2021). Emerging from customer connect to redefining the business models, DT has become all pervasive. Irrespective of the industry type, scale, or scope, DT remodels the business processes and aids in value creation at all levels. The processes that form organizations' core competencies traditionally are being

redefined. As a result, strategy coupled with technology formulates the core aspect of DT (Kane et al. 2015; Chawla et al. 2022).

The strategic importance of the IT provokes the organizations for appointment of CIOs (Chatterjee et al. 2001). As the acknowledged head of IT for the organization, the CIOs are deemed responsible for making DT a strategic business priority (Hess et al. 2020). It may be noted, however, that the position is not new and has evolved over the years (Chawla and Goyal 2021). During Mid 80's as distinguished business functions emerged, IT and IT-specific managerial roles started becoming more evident. Subsequently, managing noticeable IT function, these IS managers tussled for C-suite position. This got corporate attention, and consequently, 'CIO' as a title emerged (Chun and Mooney 2009). With technology-oriented innovations in early 1990s, the CIO role started shifting from a core technical role to a strategic one. Ever since then, the role of the CIO has been very dynamic and is continuously evolving (Davis and McDonagh 2015; Hillebrand and Westner 2021).

As evident from Table 1, the role has continuously expanded from being focused on technology and operations to include innovation and strategy (Ross and Feeny 1999; Davis and McDonagh 2015; Davison et al. 2023). This is consistent with the observation in role theory that a social actors' role continuously changes in response to a relational process (Biddle 1986; Luhmann 1995; Raes et al. 2011). In fact, in the case of CIO role, the naming itself has undergone some change. With an increased organizational orientation towards digitalization, organizations are adopting the new senior-level position of Chief Digital Officer (Friedrich et al. 2016; Péladeau et al. 2017). The popularity of the CDO position has notably increased in recent year (Kunisch et al. 2020). This has inspired researchers to investigate the developing role of the CDO. Consequently, there have been many studies representing multiple aspects of the CDO role (Becker et al. 2018; Tumbas et al. 2017; Reck and Fliander 2019; Singh et al. 2020). CDO is generally defined as a strategic management role responsible for defining digital strategy and directing organization's DT (Haffke et al. 2016; Horlacher et al., 2016; Singh and Hess 2017; Artemenco 2020; Moker 2020, Raković et al. 2022, Davison et al. 2023). This entails reorganizing a company, forming a vision, and developing a thorough plan for digitally transforming business operations (Artemenco 2020). Singh and Hess (2017) note that the CDO position is distinct from other business functional roles and is primarily focused towards bringing digital initiatives and fostering a digital-oriented culture in the organizations. Singh et al. (2020) note that the primary responsibility of the CDO is to coordinate vertically across hierarchical levels and horizontally between organizational units. They observe that the CDO role impacts multiple organizational aspects such as strategy, customer engagement, innovation, and data analytics practices. Along the same lines, Scholars (Matt et al. 2015; Horlacher and Hess 2014; Davison et al. 2023) note crucial managerial role of CDO as an in charge of the DT strategy. Singh and Hess (2017) suggest the role of CDO to be broader than CIO because the CDO is responsible for connecting with employees, understanding business processes, and so on. Further, it is suggested that the companies provide concrete guidelines for better aligning the two roles of CDO and CIO so that they can work in close collaboration (Tumbas et al. 2017; Buchwald and Lorenz 2020). For instance, Horlacher and Hess (2016) suggest that whilst the strategic and communication components of

Table 1 CIO roles outlined in existing literature (Adopted and extended from Hütter and Riedl 2017)

Authors	Defined role of CIO
Grover et al. (1993)	Entrepreneur, Leader, Liaison, Monitor, Resource Allocator, Spokesman
Ross and Feeny (1999)	Business visionary, Functional head, Strategic Partner
Smaltz et al. (2006)	Information steward, Integrator, IT educator, Relationship architect, Strategist, Utilities provider
Leidner and Mackay (2007)	Big-bang CIO, Keep-it-running CIO, Value-adding CIO
Preston et al. (2008)	IT advisor, IT laggard, IT mechanic, IT orchestrator
Chun and Mooney (2009)	Innovator and creator, Landscape cultivator, Opportunity seeker, Triage nurse and firefighter
Chen and Wu (2011)	Business visionary, Business System Thinker, Entrepreneur, Informed buyer, Infrastructure builder, Organizational designer, Relationship builder, Value configures
Peppard et al. (2011)	Agility IT director CIO, Evangelist CIO, Facilitator CIO, Innovator CIO, Utility IT director
Guillemette and Paré (2012)	Architecture builder, Partner, Project coordinator, Systems provider, Technological leader
Li et al. (2012)	Information strategist, Integrator, IS contract oversight, IS Strategist, IT manager
Weill and Woerner (2013)	Embedded CIO, Enterprise processes CIO, IT services CIO, External customer CIO
McLean & Smits (2014)	Technologist, Enabler, Innovator, and Strategist
Haffke et al. (2016)	Evangelist, Coordinator, Innovator, and Advocate
Singh and Hess (2017)	The entrepreneur CDO, The digital evangelist, The digital coordinator
Tumbas et al. (2017)	The digital accelerator, The digital marketer, The digital harmonizer
Kunisch et al. (2020)	Generalist, Specialist
Hillebrand and Westner (2021)	Technology provider, Innovation driver, Relationship manager, Integration advisor
Kratzer et al. (2022a, b)	Strategic IT manager, Restructurer, Rapid scaler, and Hands-on supporter

DT can be handled by CDOs, CIOs can handle the technical aspects. Tumbas et al. (2017) observe that the CDO role is unique, and its responsibilities can significantly vary amongst organizations (Péladeau et al., 2017). CDOs come from various backgrounds, but the most prevalent are sales and marketing, followed by information technology (Planer-Friedrich and Sahm 2015). This illustrates that the CDO role is of interest to executives from multiple organizational verticals and DT is not limited to technology. Hence, even though the CDOs might not be leading IT directly, they become core promoters of technology change in the organizations (Berman 2020; Davison et al. 2023; Dumeresque 2014, Raković 2022), acting as a bridge between IT and business (Ctarino 2018).

However, Drechsler et al. (2019) suggest that the establishment of a CDO alongside a CIO may convey a negative impression because of the likelihood of overlaps or even conflicts of power. Unambiguous responsibilities at the C-level are required to increase the enterprise capacities in times of upheaval. Clear segregation

of responsibilities is significant because when the CDO and the CIO coexist, the absence of clear responsibilities of the CDO function and profile offers some room for eventual conflicts (Ctarino et al. 2018). Seth (2021) mentions that choosing who should lead DT for the organization is a tough choice and one that often creates friction as DT cuts across business functions. Notwithstanding the title—CIO, CDO, or IT director—we define this executive as CIO representing leader crucial to strategic decision making for digitally transforming a cross functional organization (Friedrich 2016). Moreover, since our case organizations use the title of CIO to designate their IT/DT leader, taking a grounded theory approach, we stick to the term CIO.

The CIOs need a high level of technical expertise (Smaltz et al. 2006) to manage IT and to equip the organizational users with necessary IT tools. The technical competency is also required to manage IT infrastructures complexity (Sobol and Klein 2009). The CIOs carefully plan the budget and IT spending to bring business agility (Leidner et al. 2008). More importantly, in context of DT, aligning IT to business is one of the foremost tasks of the CIOs. Therefore, the CIOs must possess foundational knowledge about the business process along with technical skills (Preston et al. 2008). While acting as C-level executives, the CIOs focus on organizational strategy. At the same time, acting as technical managers, they rationalize IT infrastructure to control costs (Al-Taie et al. 2014). Peppard et al (2011) suggest that the CIO's comprehensive understanding of business processes is a must to provide business innovations. In similar context, Banker et al. (2022) argue that organization's preference of CIO skills (technical vs business acumen) is influenced by the organization's strategic positioning. This is also a signaling mechanism as CIO appointment invokes reaction from marketplace and impacts shareholder's wealth (Chatterjee et al. 2001).

For DT to be successful, scholars (Al-Taie et al. 2014; Preston et al. 2008; Loonam et al. 2014) suggest that top management team (TMT) members should be able to appreciate the value addition by digital technology. CIO's structural relationship with TMT, CFO and CEO defines the influence of CIO in strategic decision-making for the organizations (Hunter 2010; Johnson and Lederer 2010; Loonam et al. 2014). CIOs create shared vision for strategic alignment of IT strategy with business strategy (Preston and Karahanna 2009) and educate TMT about what digital technologies can offer to the business (Weill and Woerner 2013). The rise of new competitors who use disruptive business strategies frequently poses a threat (Cozzolino et al. 2018) to the organizations. Therefore, many times, even though the TMT members may have heard about competitor's technological success stories, they may not be sure how to replicate it at their business place (Leidner and Mackay 2007). In such situations, CIOs provide organization specific insights to TMT on building strategic digital capabilities (Al-Taie et al. 2014; Preston and Karahanna 2009).

With greater technological awareness, DT is grabbing organizational attention. DT entails experimenting across the organization with new age digital technologies and demands change management (Hansen et al. 2011; Hanelt et al. 2021; Zaki 2019). To achieve this, CIOs build the innovation culture and encourage employees for experimenting with digital technologies (Preston et al. 2008). They leverage their communication skills to motivate employees across the organization (Preston and

Karahanna 2009). To foster organization-wide collaborations, CIOs need to build inter-unit relationships, come out of the IT department boundaries, break organizational silos, and communicate effectively with business (Brown 1999; Grover et al. 1993).

Whilst not directly referring to CIO/CDO, based on an extensive literature review, McCarthy et al. (2021) suggest eight DT leadership characteristics as a digital strategist, digital architect, digital culturalist, organizational agilist, customer centrist, business process optimizer, digital workplace landscaper, and data advocate. Hence, the CIOs in the digital era (or, CDO) need to intelligently perform many organization-level roles simultaneously. These roles have high organizational context specificity (Al-Taie et al. 2014; Singh et al. 2020). However, despite the insights from literature, the practical tasks that CIOs should perform in the context of DT and how they do it is still not empirically established. Hence, it becomes essential to understand and reflect on the pragmatic aspects of CIO role. We explore more on this viewpoint empirically in this study. In the next section, we outline the methodology used in this study.

3 Research methodology

The objective of this research is to empirically investigate the contributions of CIOs in DT. As DT is a practice-oriented subject, we explore the phenomenon from the practitioners' perspective (Strauss and Corbin 1997) on how they contribute to DT in their specific context. Consequently, an exploratory research methodology with qualitative method was adopted. Since we wanted to explore the phenomenon in the real-world settings, we selected an inductive approach using multiple case studies (Yin 2017; Saunders et al. 2018). Representative cases from different industry segment were selected to obtain maximum information from a small set of sampled organizations from an emerging economy. Since we were more interested in organizations for which IT was not a key business but could be a critical enabler, we purposefully did not include organizations from the IT sector. Established (more than ten years of existence) organizations that are large (annual revenue more than10 millions USD) and have made some significant efforts in DT were identified. To screen sampled organizations, we used publicly available information and approached twelve organizations representing different industries. Six out of those accepted our research participation request. The number of cases are deemed fit as per Eisenhardt (1989) who suggests that inference from four to ten cases forms a reasonable basis for theory building and generalization. Table 2 presents the details of the case organizations that participated in the study. The.

It may be noted that the names of the organizations are withheld for ensuring anonymity. For data collection, a set of interview questions was prepared as an interview guide. For this study, the interview questions were prepared in two steps. First, we determined and compiled a list of the pertinent factors related to our research topic. Then, we developed all-encompassing questions that addressed those factors. In the next step, the drafted questions were discussed with a panel of academicians and industry practitioners to validate the question and to have suggestions for adding

Table 2 Profile of case organizations

S No	Organization Pseudonym	Sector/industry	Revenue (USD)	Employees (Thousands)	Number of interview participants
1	AutoCo	Automobile Ancillary	78 Mn	1–5	3
2	CementCo	Cement Manufacturing	555.2 Mn	1–5	3
3	HealthCo	Hospital	14 Mn	< 1	3
4	ChemCo	Chemical Manufacturing	1077 Mn	1–5	2
5	BuildCo	Roofing Solutions	440 Mn	1–5	3
6	TextileCo	Textile Processing	80.00 Mn	5–8	3

any additional aspect. The finalized interview questions are mentioned in Appendix 2. We used this as a baseline to conduct interviews with the CIOs, business executives and their technology partners as a source of primary data. Here it is imperative to mention that we aimed at conducting the interview with the person responsible for digital leadership irrespective of their formal title (CIO or CDO). However, in all the selected cases, this role was managed by the CIOs with no CDO position existing in the organization.

In total, 17 semi-structured interviews were conducted between April 2021 and November 2021. The average duration of the interview was 55 min. During the interviews, multiple aspects covering IT and business management practices, organizational digital initiatives, their impacts, and CIO's contribution to DT, especially from an empirical standpoint, were discussed. After conducting the interviews, the data was transcribed from audio files into text files. For analysis, qualitative coding was done using Gioia method (Gioia et al. 2013). The choice of the Gioia method for this research was motivated by two key factors. First, we recognized that it was essential to employ a method that thoroughly satisfies the requirements of qualitative data analysis. Second, we sought a systematic method that would allow us to refine the acquired data in a practical way without affecting its quality or authenticity.

Gioia Method systematically presents the first order researcher analyzed concepts known as "informant centric codes", and subsequently second-order researcher analyzed concepts known as "researcher derived themes". These second-order themes are finally distilled as the overarching dimensions, and a consolidated data structure is created, finally leading to concept building. For this research, the evaluation of the cases began after adequate data had been obtained right after the first interview. These interview transcripts were analyzed line by line, and data coding was performed following methodological guidelines by Gioia et al. (2013). The first level concepts were systematically derived from the interview statements from the data transcriptions (Corley and Gioia 2004). The quotes presenting similar contexts were clubbed and sorted together to present common concepts. At first, all the interviews were looked at one at a time, and then cross-case themes analysis was used to look at the data through multiple lenses. Appendix 1 shows some representative examples. After attaining the saturation point (Corbin and Strauss 1990; Pabst von Ohain 2019), where no more codes could be produced, the focus shifted to the second

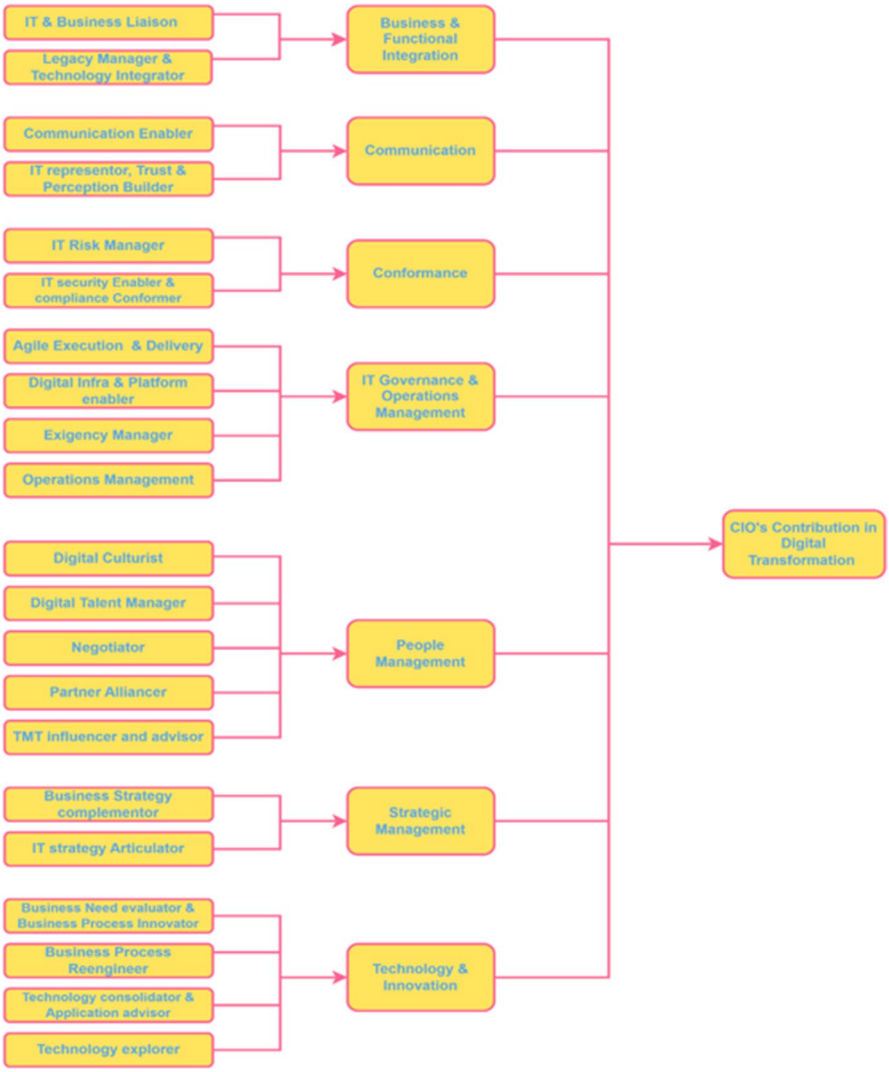

Fig. 1 Coding tree based on Gioia method

stage, which involved developing second-order constructs. In the next iteration, the second-order constructs were grouped under more manageable and meaningful constructs (Gulati and Reaiche 2020). The constructs so formulated were termed as overarching dimensions, reflecting CIO role. The dimensions thus identified are shown in Fig. 1 and are presented in the next section. The arrows in Fig. 1 depict the linkage of informant centric codes to the respective derived (second-order) themes. It may be noted that the linkage between informant centric code and role is based upon best fit and does not imply exclusivity. Similarly, the roles identified are not straight jacketed and can be inter-dependent. For instance, communication is a key

requirement in people management, or business and functional integration is closely entwined with technology and innovation.

4 Findings

This section presents the contributions of CIOs and derived roles in context of DT. These insights are primarily derived from our discussion with interview participants and thus empirically outline CIO role in DT.

4.1 Business and functional integration

4.1.1 IT and business liaison

DT is an organization-wide phenomenon that strategically covers everyone in the organization. Therefore, unlike traditional IT management, DT demands close integration of IT with business. As the functional head of IT, the CIO looks at the big picture, holistically analyzes business demands and ensures collaborative interactions of business functions. The CementCo CIO notes, *"For DT to be successful, we have to understand all the nuances, the business impacts, its feasibility and its executional perspective [..]. All has to be understood."* The CIOs act as a functional integrator and educators to businesses to institute the IT solutions for the business goals. Through continuous interaction with business, the CIOs evaluate what information is essential, assess its delivery plan and bring greater alignment between business and IT to foster productivity for the organizations. As the AutoCo CIO notes, *"Every department approached us individually with its requirements without any holistic organizational view. [..], I discussed with the executive board and proposed forming a digital council. Now we have an organization-wide standardized methodology to prioritize our initiatives strategically."*

4.1.2 Legacy manager and technology integrator

Business operations run on a technology backbone with custom-designed legacy applications. These legacy applications must be kept functional for continued business operations. The CIOs stand as a custodian of legacy systems to support smooth running of business operations. As the CementCo CIO mentions, "*My first task for the day is to ensure everything is up and running*." At times, new applications depend on the data generated by these legacy applications. Therefore, legacy applications' integration with digital technologies has become inevitable. The TextileCo CIO explains, *"For enabling technology and bringing digital processes in practice, the legacy was required to be integrated to the cloud."* CIOs carefully orchestrate IT architecture in terms of design compatibility and how these applications interface. The AutoCo CIO recalls, *"We developed 50 to 60 applications on a common digital platform and extended insights of our working applications."*.

4.2 Communication

4.2.1 Communication enabler

Vibrant and authentic communication is the foremost requirement to foster DT. The CIOs communicate up, down and across their levels. In the upward communication with chief executive officer (CEO)/Board members, CIOs prominently focus on projects updates and support. They appraise them about digital reforms in the organization and the status of various such initiatives. In terms of horizontal communication, the CIOs focus on bringing more coordination with business teams, by discussing innovative ideas and communicating the benefits of new digital initiatives. Communicating down the line, CIOs communicate with multiple business users for understanding their requirements, to appraise them on recent digital trends and ongoing initiatives. For this purpose, the case CIOs use media such as JAM pages, emails, newsletters, or social media channels. One AutoCo business executive explains, "*the CIO ensures that everyone is enabled to use the tools while working from home, he created a JAM page on internal social media platform to enable collaborative communication"*. in TextileCo, the CIO deployed a new social media channel to engage with the customers and spearheaded the development of a dashboard to analyze their feedback. Similarly, in CementCo, the CIO listened effectively to the sales team and catered to their request for designing a common e-meeting platform to facilitate collaboration among the field sales force, dealers, and customers.

4.2.2 IT representor, trust and perception builder

The CIOs consistently ensure that the IT team realistically commits business deliverables and adheres to the timelines as the breach of timelines may violate trust among groups. This is further synergized by formulating cross-functional teams, bringing IT and functions closer to foster business contributions. As BuildCo CIO mentions, *"When we do these cross functional collaborations, my team members come back with lots of understanding on what business is suffering from and with a lot of empathy, these initiatives ease to brining results"*. With increased digitalization, the users have also become tech-savvy. In many aspects, rather than asking, they prefer to guide the IT department about the solutions that will suit their requirement. For such discussions, CIOs orchestrate the culture of trust among the users and IT to enable bidirectional communication. As the CementCo CIO mentions, *"I guide my team members to be more receptive to business and putting their thoughts on how to complement business better?"* The CIOs represent themselves as the face of IT in various business meetings. By presenting analysis-based facts and convincing their business peers on how digital technologies can bring business success, they attempt to build a cordial perception of IT. Once a positive perception is built, it helps the CIO and the IT team in drawing support from business. The CementCo CIOs mentions about his projects, *"Most of the time they approve because it is coming from me[..] they know if it is coming from me, it must be trustworthy"*.

4.3 Conformance

4.3.1 IT risk manager

As organizations embrace DT, their interactions with multiple vendors and reliance on multiple infrastructure models increases manifold. This rapid digitalization of business poses a wide variety of critical IT risks which emerge from IT yet have organization-wide impact. CIOs, being the head of IT, are considered accountable to mitigate these critical risks that, if unmitigated, may result in hefty penalties and loss of organization's reputation. CIOs formulate policies and procedures to mitigate risk of regulatory compliances such as application licenses, mode of deployment and secure connectivity requirements. As the AutoCo CIO mentions, *"I have exclusively deployed one of by team member for licensing and risk management. He dedicatedly takes care of timely renewals and compliances. It is important [in] keeping tight controls over the systems"*.

Along with the risk mitigation, keeping the digital applications up and running is equally critical. For this, the CIOs formulate business continuity plans and set up disaster recovery (DR) sites for business continuity. As the Healtchco CIO elaborates, *"we have just set up a new DR site for all critical applications with a target RTO (recovery time objective) of four hours"*. Thus, in the context of DT, the CIOs also act as risk managers. As the BuildCo CIO explains, *"Digital Technologies change and they change at a breakneck pace. We have to ensure both forward and reverse compatibility of the systems, and it is crucial [..that] systems should be operational seamlessly and securely, today and tomorrow."*

4.3.2 IT security enabler and compliance conformer

DT enables organizations to operate beyond traditional organizational boundaries. In so doing, it adds more exposed points with increased risk of compromising data. Therefore, access management and data security become paramount when designing digital applications. As each business process is unique, finding system vulnerabilities demands in-depth knowledge of entire business operations. With their thorough understanding of business applications, technologies, and IT infrastructure, CIOs are best placed to holistically assess security threats and keeping the systems secured. As the CementCo. CIO mentions, *"I keep auditing the network to expose vulnerabilities to ensure applications are secure and the possible intrusion points are appropriately monitored."* CIOs set appropriate IT security standards such as ITGC (IT General Controls) in consultation with relevant authorities. In the long run, such IT compliance requirements also become a part of the integrated organizational compliance. For example, In the HealthCo, the CIO mentioned adherence to the guidelines issued by the government for keeping patients' data secure. To ensure conformance to these set standards, CIOs establish and govern internal IT processes such as controlling super-user and administrator accesses, formulating password policies, setting up rules for data backups, cross-application segregation of duties, and conducting process audits. As the CementCo CIO notes, *"We have set up demilitarized zones for secure operations, handshaking and data transfers, with*

our external partners, wherever data sharing has to happen on day-to-day basis." The CIOs further involve external audit agencies for detective and automatic control audits to close the potential gaps before the applications were launched for business use.

4.4 IT governance and operations management

4.4.1 Agile execution and delivery

Digitalization has made business cycles shorter, enabling quicker launch of new products and offerings. Following agile practices, organizations now strategize the product development iteratively and frequently. To keep the agile pace of implementation, CIOs set up the IT functional processes in a way that best suits the business pace of development. They direct their teams with actionable insights and set up mechanisms to get early warnings. As the ChemCo CIO observes, *"we followed methodology which is not exactly agile, but a mix of agile and waterfall. We do daily scrums and build iteratively but at the same time we take care of business pace of adoption and adjust our development pace accordingly"*. Further, we noticed that specifically for external support, our case CIOs experiment a lot with startups as they believe that the startups are much more agile than their established vendors. One of the technology partner consultants of CementCo observes, *"Their intent is to involve more and more startups[..] as they expect them to bring fresh ideas and agility alongside cost effectiveness"*. As DT involves experimenting with technology, not every initiative is expected to be successful. CIOs encourage piloting practices to promote innovations at business. They look at past failures, improvise, and redevelop. This brings synergies in both cost and schedule. As TextileCo CIO notes, *"Failing fast and scale rapidly is our mantra"*.

4.4.2 Digital infrastructure and platform enabler

The success of DT depends heavily on organization's infrastructure capabilities. Digital technologies demand infrastructure that is easily accessible and rapidly scalable to offer deployment of new components. The CIOs are tasked to set up the appropriate infrastructures, including data center and disaster recovery backup. Since DT demands a high number of applications, the task of planning and managing infrastructure has become extraordinarily complex. As AutoCo CIO elaborates, "*The basic core of DT is data…. ever growing enormous data. We have to incisively design infra considering sizing, costing, uptime and our landscape's compatibility for integration, whether on cloud or on-premise*.".Usually, our case CIOs start with building infrastructural architecture for establishing DT. Application deployment on cloud or on-premise data centres becomes a key strategic decision for the CIOs. As BuildCo CIO observes, *"Infrastructure advancements and cloud-based model adoptions were our priority for all digital applications developments and their backups."* Moreover, organizations are now moving to a data-driven business model centered around analytics and business intelligence. Though DT requires multiple

applications, insights from these applications need to be cohesive. Such consolidation must happen at the infrastructure level to enable the interlinking and data sharing. The BuildCo. CIO mentions, "*The executives demand a consolidated dashboard. Applications, empowered by robust and reliable infra, must speak to each other*."

4.4.3 Exigency manager

An uncertain business environment may result in situations requiring a quick response. These exigencies mostly emerge from the external environment. If left unmanaged, such exigencies would result in the loss of business opportunities or create disruptions to the business. Many times, these exigencies are beyond technology. Hence, these demand close collaborations of business teams with IT. With their business know-how, personal relations, and technological acumen, the CIOs manage these situations with dexterity. As the AutoCo CIO recalls, "…*Suddenly, the shortage of semiconductor chips impacted our production schedules; raw material supplies had to be replanned. We had to quickly build an app for collaborative planning and faster communications*." Almost all CIOs we interviewed mention that the Covid pandemic brought high turbulence to businesses requiring immediate move to remote working. However, it seemed to impact HealthCo the most, wherein the CIO transformed this business necessity into a profitable revenue model for the organization by digitalizing entire processes. HealthCo CIO recalls, *"We digitized the prints, reports, X-ray films… and launched tele-consulting app with doctors, it was critical as our objective was to reduce the hospital visits of patients and at the same time providing healthcare to every patient"*. The AutoCo CIO recalls another exigency, *"The management decided for another joint venture, usually this is disclosed after all the formalities are done, but we had to do technology integration, understand their systems, processes, resources, but time was a constraint."* In summary, business exigencies emerge from time to time, and CIOs are responsible for managing these exigencies, often with limited resources under very stringent timelines. Hence, CIOs in this role mostly act as firefighters to business.

4.4.4 Operations management

Digital Operations Management is the pivot to digital business operations and CIOs are considered responsible for smooth operations of IT applications. It becomes crucial to oversee the IT operations across all functional domains and to ensure that users get a consistent IT support. All our case CIOs established a central IT service desk to address IT support governed by predefined service level agreements. As one business user in ChemCo commented, "*With IT issues, the business users were not sure whom to get in touch with for which application, the support was person specific [..] but now they (IT) have set up a central helpline, it is convenient for us (users) to get the issues logged and trace the status"*. As the DT technologies are new, their deployment is also unique in many technical aspects. CIOs designate their teams to monitor critical parameter. As digital applications prominently need the internet to work, bandwidth management becomes a critical operational

concern. The CIOs advise their network team to reassess the bandwidth frequently. As AutoCo CIO observes, *"seamless connectivity is important for the uniform experience of users and even for the office visitors. With more digital applications on board, we have upgraded our bandwidth at regional offices"*.

4.5 People management

4.5.1 Digital culturist

One of the prerequisites for DT to be successful is that the organizations must adopt to novel ways of working. As the BuildCo CIO observes, "*We have a diversified workforce and many diversified work profiles. But we want everyone to adopt, learn and contribute the digital way; we need to foster digital learning innovatively to make it mainstream thinking"*. CIOs enable faster adoption of new initiatives by presenting the big picture of how the changes will add value to the organization and employees. Further, they facilitate adoption through novel ways such as e-trainings, self-service Kiosks, and online facilities. They further devise new technology-oriented ways of employee engagement and monitoring employee activities, measuring them digitally and appropriately setting up feedback mechanisms. The HealthCo CIO recalls, "*We brought in the concept of digital prescription, the doctors were not ready to adopt it as they were used to pen and paper, they sought excuses such as time constraints, but ultimately we convinced them"*. For successful DT, CIOs keep experimenting with new digital technologies to identify ones that best fit the organizational requirements. They foster collaborations with new technology partners, especially startups, to bring innovative thinking. AutoCo CIO observes, *"It is important to continue learning, learn what is new, how that can fit in our context and each of my team member must explore on that [..] we fail fast and we learn from it"*.

4.5.2 Talent manager

Since digital technologies are always in demand, talent retention a pertinent issue for CIOs, more so in DT context. ChemCo CIO recalls, *"We recruited two technical consultants for big data dashboards, they started picking up but then suddenly moved on. We had similar experience with other technology resources also.. Stability of workforce is important"*. Talent management entails assessing the talent demand, deciding how to fill talent gaps, recruiting the right talent, channelizing them in the right direction and training them from time to time. In all our cases, we observed that the CIOs actively managed the talent for DT as per their organizational fitment. For instance, the CementCo CIO came up with the innovative idea of training business users as the core team rather than recruiting full-time technology consultants. He explains the rationale, *"The new digital skill to the business users will complement their routine business activities, they can better think of more value additions with same technology, they will train others and at the same time their chances of retention would be more"*. In the HealthCo, the CIO kept the IT team members as the core team responsible for DT while complementing them with a

remote handholding team from a technology partner to interact on a need basis. In contrast, BuildCo CIO chose to recruit new talent for building internal digital competency for designing the dashboards. At the same time, CIOs give utmost importance to talent retention. For this, CIOs build strong connections with their teams to understand and address the ground-level issues, which otherwise would result in high attritions and low morale. They inspire team members to innovate and deliver. As the TextileCo CIO reveals, "*I keep myself always available for my team; A friendly environment brings better synergies to the workplace*."

4.5.3 Negotiator

An essential aspect of DT is to have the right technology with the proper budget. With increased focus on digital, CIOs need funds to cater to business demands. To achieve this, they need to convince a lot of stakeholders, including the top management team (TMT) and the board. The CIOs thoroughly analyze to choose digital projects with better ROI (Returns-on-Investment). As the AutoCo CIO narrates, "*I have to deal with many internal and external stakeholders and mould the things in favor, be it promoting new initiatives, delivery timelines, budget or resources.*" Unlike usual business negotiations, the technology product and service purchases require a higher level of information. With their technological background, CIOs best understand the technical requirements to negotiate deals with technology providers. HealthCo CIO explains, *"we open new relationships by listening to them (vendors) carefully and mentioning the big collaborative opportunities they are looking for"*. CIOs analyze the big picture well before negotiation. This means that they holistically analyze and consolidate costs for all digital business requirements such as infrastructure, deployment, development, and training. They then negotiate in totality to get an advantage of economies of scale. CIOs offer their organizational collaborations to vendors for technology pilots and act as their reference customer to make it a mutual advantage. TextileCo CIO mentions, *"Without being anxious, I discuss the big picture, evaluate how the relationship can be sustainably benefiting both parties"*. Within the organizations, CIOs negotiate with employees for digital adoption. With a more digital-savvy workforce, the CIOs negotiate accepted solutions that fit in budgetary constraints. As the ChemCo CIO comments, "*The digital-savvy workforce has high-end technological workplace expectations while the old gen has adaptability challenges; I have to keep a balanced approach to cater to both*".

4.5.4 Partner alliancer

For DT, success depends on the synergies generated from collaboration. Hence, the CIO establish strategic relationships with partners to build innovative business solutions. As the AutoCo CIO mentions, *"We have strong collaborations with technology partners who keep us updated on their offerings and* vice versa*. Their technical teams are aware of our landscape."* They also leverage their associations with partners and governing bodies to develop their understanding of technological advancements. HealthCo CIO notes, *"I keep attending the conferences organized by the*

technology partners time to time. These give me a fairly good inclination of what is being offered and what others are doing in the ecosystem". CIOs promote customer collaborations for new digital product designs, effective delivery of services and build sustainable relationships. To the customer side, they evaluate customers' acceptance and effectiveness of the offered digital solutions, which ultimately supports the marketing team to price and position new products. As the HealthCo CIO mentions, "*I round the floor, meet patients and understand their expectations from the technology perspective; I get real time feedback*." To the vendor side, they devise mutually win–win situations such as time and material contracts, ticketing-based support, offering digital platforms as trials, deployment of new products. These relationships indirectly support organizations to bargain knowledgeable resources and competitive bids from vendors.

4.5.5 TMT influencer and advisor

The CIOs act as a trusted technology advisors to the TMT. They bring awareness to TMT members about which areas are critical for the organization to pursue DT to take maximum benefit and how that should be structured. In various communications to the board, the CIOs present competitive insights on which digital technologies the competitors are working on. HealthCo CIO notes, *"I keep TMT members updated on the recent digital technological trends with the intent that they may find them of some novel use at our business"*. The CIOs develop an understanding of TMT's requirement and functionally narrate this requirement to their teams for developing insightful dashboards to present real time insights to TMT. This way, they bridge the knowledge gap and empower the TMT to make data-driven decisions with agility. The CIOs also inform the TMT on the status of projects they approved. The CIOs presents an accurate picture of the ongoing issues and concerns at ground level to TMT to allow for timely actions, which otherwise would result in low organizational productivity. As the ChemCo CIO comments, "*The technological awareness to senior executives is not very high, sustained discussions are important to keep them appraised of digital trends*."

4.6 Strategic management

4.6.1 Business strategy complementor

A key task of CIOs in the context of DT is to complement the business strategy from a digital perspective. In doing so, CIOs usually participates in the strategic business planning meetings to understand the business vision and to formulate business plans to operate in the next couple of years. CementCo CIO comments, *"Once a year we have a strategic meet, where in we collectively discuss business vision and set the yearly objectives. Then we have quarterly meet to assess our performance and plan for subsequent quarter"*. After discussing with the CEO, CFO, and other TMT members, the CIOs design the digital roadmap. AutoCo CIO explains, *"TMT members are mostly aware of business trends, they are quite brainy. I deliberate and*

plan the digital strategy with them". They clearly articulate the business touchpoints where digital technologies can add business value. More specifically, the CIOs suggest function-wise value additions such as streamlining the supply chains, collaborations with partners, digitally enabled customer-oriented features for product development, digitally oriented new service models, business process reengineering and robotic process automation of internal business processes.

4.6.2 Digital strategy articulator

CIOs put in their leadership insights to plan the future of digitally enabled businesses. With their techno-functional expertise, they articulate how IT will functionally achieve the business objective through digital technologies. They formulate digital vision and digital roadmap with designated timelines to set the agenda for translating the digital strategy into attainable goals. Next, while setting the objectives for their teams, they plan support mechanisms such as training and expert guidance to facilitate their teams in achieving these goals. As the TextileCo CIO mentions, "*I have to aggregate the bits and pieces, shape them, and bring them to ground level attainable tasks that follow the team's planning for delivery*." At the same time, CIOs perform strategic decisions such as make vs buy, delivery from internal teams or partners engagement, cloud vs on-premise deployment, infrastructure refreshes and on IT security aspects. They critically rationalize the spending year on year and carefully plan IT spending within allocated budget. HealthCo CIO notes, "*We purchased infra last year to set up our new data center, we now have to capitalize that and increase spending on digital applications development"*. Similarly, the CementCo CIO signed a strategic collaboration, first in the country, with one of the topmost technology partners for business big data analytics application development.

4.7 Technology and innovation

4.7.1 Business need evaluator and business process innovator

The CIOs aim to empower every process and person at the organization for working efficiently with technological tools. They evaluate the pain points of business to find out areas of improvement and how digital technologies can bring change across them. All our case CIOs resonated with the importance of innovation using digital technologies. CementCo CIO mentions, *"I always focus on what new can be done with technology, how do I simplify the business, bring revenue to business"*. Similarly, the AutoCo CIO mentioned the implementation of success factors for human resource process innovation. He recalls, *"We wanted to bring innovation to HR function [..] with Integrated data in Success Factors, we can get educational qualification, experience, and interest areas of our employees. Using this innovative tool, we place them in relevant roles"*. HealthCo CIO mentioned automating IT support with implementation of a new artificial intelligence based ticket automation tool for which an HealthCo executive comments, *"It is easier to get first line support now,*

the chatbot suggests the resolution, logs the ticket also and proposes which team to get in touch along with their contact details."

4.7.2 Business process reengineer

CIOs look at the opportunities to transform business processes end-to-end. In so doing, they first minutely analyze the business process workflows. Thereafter, they assemble and govern the teams to reengineer the business processes in close collaboration with business executives, to bring radical improvements in productivity and efficiency. TextileCo CIO recalls, *"The financial reconciliation of invoices was a key concern of finance department at the month ends; we implemented an RPA (robotic process automation) tool to completely automate the reconciliation process. It processes invoices every day after business hours. Any errors are mailed automatically to business executives"*. Similarly, the CementCo CIO received a concern of salespeople not been able to physically meet with sales dealers due to pandemic restrictions. The CIO customized a new mobility app for virtually connecting with dealers. As he puts it, *"Earlier sales managers could do 4 to 5 dealers a day, now with customer connect app they do 12 meetings per day. The solution has brought synergies across sales process"*.

4.7.3 Technology consolidator and application advisor

Business functions often tend to work in silos. This siloed structure gradually results in functions getting trapped in bounded rationality. Moreover, this phenomenon is not just limited to business processes; technology also becomes a part of it. Working in silos, the business departments also choose niche technologies for their department functions. This results in multiple technologies, inter-departmental conflicts and multiple applications being used at the same time posing practical difficulties such as tracking and annual maintenance contracts renewals. Therefore, CIOs critically examine the need of onboarding of any new technology. CementCo CIO mentions, *"we cannot manage too many things, we need to consolidate our technological landscape"*. The CIOs attempt to replace these redundant applications with universally applicable digital technologies to bring uniformity across the organization. AutoCo CIO recalls, *"we rationalized the business requirements, consolidated applications and migrated them on new single digital platform, we eliminated some thirty to forty redundant applications"*. Further, as an extension to the application consolidator role, we observed another phenomenon where the business users often encounter the functional issues for which they are either not aware of technological solutions or unable to decide on suitable technology. In such cases, the CIOs, acting as application advisors, critically assess the business requirement and suggest suitable solutions. They formulate the technological design architecture and make sure that appropriate technologies should be in place.

4.7.4 Technology explorer

To keep abreast of technological developments and to bring new technological solutions to their organizations, the CIOs explore the technology landscape continuously. Their professional network, technology partners and customers become the critical source of information for them. The CIOs build strategic partnerships with the governing bodies, take professional memberships, build relationships with reputed academic institutions to keep themselves abreast with the recent developments. The TextileCo CIO shares, *"I keep attending various events organized by the technology partners. I met other CIOs there. This keeps me updated on what is happening around"*. Further, the CIOs spend time with customers to build an understanding of their expectations from the product. This gives CIOs an inclination about which digital technology and application can be used to keep customers engaged. In all the cases, we observed that the CIOs spend significant time exploring the external technological environment.

5 Discussion

Traditionally evolved from a technological channel for engaging customers to a modern-day organizational strategy, DT has become one of the most critical corporate megatrends. Digital Technologies are distinctively impacting almost every business aspect. Specifically, customer demands are changing, product lifecycles are becoming smaller, data is becoming the core of decision making. This change beyond technology has proliferated to strategic dimensions of modern-day business organizations, demanding a complete revamp of legacy processes to new agile and flexible digitally transformed business models. Like other business domains, this nascent phenomenon of DT has radically impacted the expectations of the leaders, who are now expected to bring change at multiple levels such as strategy, technology, business processes, talent, product line and culture (Hess et al. 2020; Kane et al. 2015; Nambisan et al. 2019; Hillebrand and Westner 2021; Verhoef et al. 2021; Vial 2019; Davison et al. 2023). This multidimensionality of DT, in practice, creates confusion at the corporate level and therefore delivering DT initiatives in practice is not easy.

Our findings suggest that all our case organizations have realized the importance of DT and have made DT initiatives their strategic priority. A plethora of factors such as competitive pressure, customer demands, employee expectations, and legal compliances to pandemic push have accelerated the need for DT (Cozzolino et al. 2018; Verhoef et al. 2021). We also find that The CIOs are being relied upon to bring digital-oriented transformations into the business[2] (Singh and Hess 2017). This supports the observations that skilled and competent leadership is most influential for successful DT (McCarthy et al. 2021). The strategic power of IT is becoming

[2] Since all organizations had CIOs leading the DT effort, the discussion is centered around CIO. However, similar responsibilities would apply to CDO if they were leading DT.

more critical in the DT era (Bharadwaj et al. 2013), adding additional dimensions to the role of the CIO. Since many of these dimensions are competing and conflicting (Al-Taie et al. 2014; Chun and Mooney 2009; Seth 2021), it has extended the role of CIOs from department managers of a nonvalue-adding function to business-level negotiators and strategists (Bharadwaj et al. 2013; Whittington 2019). This was quite visible in all our case organizations where the CIOs were leveraging the digital technologies for transforming the business processes and simultaneously formulating the forward-looking digital business strategy. Thus, CIOs in the DT era need to master both supply and demand-side leadership aspects (Chen et al. 2010).

It is incredibly critical to depict a clear picture of what the future business will look like. Therefore, the CIOs need to be aware of how the fundamental aspects of the business are changing. Our case CIOs leveraged insights from TMT to orchestrate and align digital strategy with business (Banker et al. 2011; Feng et al. 2021). To contribute strategically, CIOs access to TMT is essential (Davision 2023). All our case CIOs had bidirectional and closely synchronized interactions with TMT members. These relationships with TMT brought tactical business information to CIOs, leveraging which the CIOs formulated the digital strategy to shape organization's digital future. All our case CIOs were part of strategic business planning and executive steering committee meetings where they shared the digital vision with other executives. They leveraged these meetings to accord the buy-in and support of executives to achieve success in the DT initiatives. The observations suggest that CIOs attain structural power from TMT interactions, and these interactions influence the strategic contribution of CIO's (Grover et al. 1993; Loonam et al. 2014; Preston et al. 2008; Preston and Karahanna 2009). This was quite evident in the AutoCo and CementCo, where these CIOs attained more importance than CIOs in other case companies and successfully delivered many digital initiatives. In contrast, the HealthCo CIO had a relatively low level of TMT involvement, and leading DT initiatives posed more challenges to the CIO. This is in line with observations that TMT should be aware of and active participation in DT (Loonam et al. 2014; Weill and Woerner 2013; Feng et al. 2021) and that CIO role is contextual on how business views IT (Al-Taie et al. 2014).

While existing literature notes the importance of business-IT alignment at the strategic level, our findings also underscore that CIOs should have a thorough knowledge of business processes to enable DT (Peppard et al. 2011; Preston et al. 2008; Smaltz et al. 2006; Capitani 2018). All our case CIOs attained a strong understanding of business processes beyond foundational knowledge. Based on an in-depth understanding of processes and prevailing practices, they could accurately pinpoint where digitalization would be most effective for business. At the same time, CIOs also possess solid technical competencies to lead and govern digital initiatives (Leidner et al. 2008; Smaltz et al. 2006; Sobol and Klein 2009; Hütter and Riedl 2017). We observed that CIOs established robust governance mechanisms to support business-as-usual operations and took complete ownership of technological aspects in operations. The CIOs emphasized decentralized IT governance to turn IT projects into business projects (Weil and Woerner 2013).

Although DT is multidimensional, covering a whole gamut of business aspects, technology still forms a significant characteristic of DT. To be successful,

organizations need to build their high-speed decision-making capabilities using key technologies. We observed that the CIOs were actively involved in initiatives such as data analytics, machine learning, and artificial intelligence for optimum value creation (Verhoef et al. 2021; Vial 2019). Specifically, the CIOs carefully considered integrational aspects to realize the full potential of digital technological designs. The AutoCo and Textile CIOs designed new applications fully integrated with legacy systems, permitting bidirectional data flow for a better omnichannel customer experience. Further, with rising customer's expectations, digital technologies are increasingly becoming part of products themselves, blurring the lines between product and services (Porter and Hapellenmann 2014). As a result, in CementCo and HealthCo, CIOs attempted to build direct connect with customers and technology partners to rapidly innovate and deliver for the business. Our case CIOs were also more inclined towards cloud adoption (Qi et al. 2022). Beyond the flexible and scalable infrastructure, the cloud offers many prebuild applications for business needs, saving both time and effort of deployment. However, this also poses decision making challenges to CIOs where they have to carefully make the trade-off decisions, such as hosting on cloud or on-premises and evaluating absolutely new types of service level agreements with their partners.

Digital technologies' ability to alter conventional competition logic encourages the development of unique management and organizational alignment techniques (Kindermann 2021). The CIOs (BuildCo, TextileCo and HealthCo) face dual challenges of fostering continuous innovation, yet with, cost reduction. This implies that CIOs must bring speedy transformation at a low cost. However, as DT entails niche skills, the resources are costly and talent attrition is high, making talent management a strategic priority for DT (Verhoef et al. 2021). The CIOs address this gap by developing cutting edge skills within their team. For this, they actively work upon upskilling their team members to facilitate high quality yet cost-effective talent. All the CIOs demonstrated a high level of mentorship by clearly defining and communicating the career path, organizing trainings, and conducting technological webinars for their team members. Further, they build personal connections with their team members to inculcate confidence in the team and keep their morale high. They also work alongside the HR function to deploy state-of-the-art tools for advanced talent retention and recruitment practices. Wherever the talent gap can not be bridged from internal organizational resources, the CIOs complemente it with partners resources. In so doing, CIOs benefit from their interpersonal skills and act as a relationship architect (Smaltz et al. 2006).

DT demands experimenting with technology to set up new ways of doing business. For CIOs, this necessitates adoption of agile practices (Verhoef et al. 2021). Our case CIOs promoted agile adoption by providing digital platforms and indulging with startups. Furthermore, all our case CIOs seem to clearly understand the importance of cultural adoption that becomes the key to the success of DT (Hess et al. 2020; Kane et al. 2015; Davison et al. 2023). To foster a digital-oriented culture, they encourage their teams and other employees for frequent experimentation with digital technologies. The CIOs promote agile project implementation approaches for building in iterations and scaling up faster to mitigate the risk of failure.

Since DT is a collaborative initiative, trust among business functions becomes a significant factor in success. Greater collaborations between IT and business are required to bring better synergies among these functions (Tan and Gallupe 2006; Peladeau 2017). To build trust and ensure commitment at all levels, CIOs need to win the confidence of the stakeholders, requiring political activity. Consistent with the recommendations found in the literature, our case CIOs engage in regular communication with the stakeholders (Enns et al. 2003; Brown 1999; Grover et al. 1993; Hütter and Riedl 2017; Davison et al. 2023). CIOs built trust with peers by setting up weekly department-level meetings to share the status of ongoing digital initiatives, address operational issues, and discuss new initiatives. They come out of departmental boundaries to break the organizational silos and set up cross-functional teams to connect with business users (Peppard et al. 2011). The CIOs act as conflict resolvers by vindicating inter-departmental conflicts through technology landscape consolidation. Consistent with the observations made by scholars (Loonam et al. 2014; Georgakakis et al. 2022), business executives were also keen to be a part of IT governance as they too realize that strategic business opportunities can not be availed in isolation without IT collaborations. Taking benefit of this mutual understanding, CIOs establish robust governance mechanisms and win necessary business support by captivating the backing of CEOs (Hunter 2010; Johnson and Lederer 2010; Feng et al. 2021). We observed that TextileCo, CementCo, and AutoCo CIOs delivered key communications through the top executive (Managing Director) to make it more influential.

Sebastian et al. (2020a, b) suggest that big old companies can bring in DT by focusing on two elements—an infrastructural backbone and a digital services platform. To provide the operational backbone, our case CIOs concentrate on establishing an Enterprise Architecture (Babar and Yu 2015). Moreover, they focus on incorporating agility in terms of technical infrastructure as well as in operational processes as agility is considered one of the key ingredients of DT (Verhoef et al. 2021). For the same reason, the CIOs from our case organizations were considering the cloud deployment of digital applications since it offers better uptime and scalability, easing infrastructural aspects for CIOs. While DT literature suggests getting rid of legacy systems, it is often difficult for traditional companies to let go of their trusted legacy systems (Sebastian et al. 2020a, b) due to various reasons such as operational continuity, sectoral context, and employee resistance. Hence, often it becomes imperative for CIO to integrate DT initiatives with the legacy systems as a strategic trade-off (Hillebrand and Westner 2021). In two cases (AutoCo and CementCo), CIOs had already migrated the legacy applications to the cloud to get away from the ageing infrastructure. To cater to the need of digital services platforms (Sebastian et al. 2020a, b; Verhoef et al. 2021), our case CIOs established new communication channels as efficient connecting mechanisms within and outside the organization.

Data privacy and cybersecurity concerns have significant impact on organization's market reputation, customer relationship, lack of trust, and social media perception (Trottier 2016; Nambisan et al. 2019). In addition to empirically demonstrating how CIOs fulfil diverse roles in the context of DT, one major contribution of this study that is not adequately covered in earlier research is highlighting the absolute

importance of cybersecurity and risk management for DT. All our case CIOs agreed that insecure digital applications can lead to potential cybersecurity and data privacy issues causing monitory and reputational damages to the organizations (Fitzgerald 2007). Digital applications generate and share a lot of data within and outside the organization. This aspect of DT makes it prone to potential IT security vulnerabilities, making it a core attention point for CIOs. All our case organizations had an IT security manager positioned under the CIO, responsible for security compliances. Our case CIOs regularly directed their teams for compliances such as updating the latest patch level, updating firewall settings, user access management and, conducting external audits for all the new applications before they were deployed on production environment for business use. The AutoCo CIO implemented governance risk and compliance tools to start tool-based authorization controls. Similarly, the HealthCo CIO also implemented secured application access and two-factor authentication systems to ensure authorized access to digital applications and further was planning to implement ISO 27001 to comply with system standards.

The issue of risk management, although somewhat covered independently in the context of the CIO role, is only recently mentioned in the context of DT (Romero et al. 2019; Menzefricke et al. 2021). Digital applications entail high risk due to numerous reasons such as scope creep, design compatibility issues, shifting business demands, infrastructural issues, and high cost of development and maintenance. Our case CIOs have devised innovative application design and deployment mechanisms to mitigate these risks. They adopt novel ways such as promoting engagements with startups (being cost-effective and agile), arranging technology partner's visits to demonstrate the digital offerings to functional heads, and prototyping of functionality with low threshold costs. They also devise innovative contracting models (such as percentage of saved cost-sharing) and negotiate extended post-implementation support with their technology partners.

To summarize the discussion above, the digital era puts leaders to the test every day. Established business processes become inflexible over time and pose a bottleneck to business processes. This also poses new challenges for CIOs who are now expected to lead, collaborate, adapt, inspire, and find ways within and outside their administrative boundaries (Davis and McDonagh 2015; Hillebrand and Westner 2021; Davison et al. 2023). The transformation from manager to leader has become imperative for the CIOs in the digital era. While traditional CIO roles (e.g., strategic alignment, IT governance, or people management) outlined in the literature are still relevant, the DT context adds some new responsibilities and nuances to the CIO role. For instance, due to the ever-expanding gamut of technology (e.g., cloud, IoT, analytics, to name a few), the issue of the technology landscape and enterprise architecture has taken precedence for CIOs. As these new technologies are still emerging and are often supplied by different vendors, the CIOs need to focus on collaboration and value co-creation. Last but not the least, since DT results in closer integration within the organization and with external partners, risk management and IT security have become extremely crucial aspects of CIO responsibilities (Fitzgerald 2007).

Considering that CIOs and other executives undertake several different roles at once while executing strategic leadership responsibilities (Cannella and Holcomb 2005), we view the functionalist, social-interactionism, and structuralism roles

(Biddle 2013; Georgakakis et al. 2022) simultaneously existing for CIOs. The core tenet of the functionalism approach is that the individual's responsibilities are dictated by the formal position or functional role that they hold in the social system. For the CIOs, the traditional IT support functions are typically presumed based on their functional titles. Alongside, the CIO's position is gaining traction in terms of DT-related functions as a part of their core responsibility. Social-interactionism perspective highlights the dynamic creation of roles by examining relational characteristics including emotions, motivation, trust, and identity (Stryker and Serpe 1982; Singh et al. 2020). This perspective prevailed to a significant extent for our case CIOs. The socio-interactionism roles evolve based on internal organizational circumstances (e.g., internal organizational ambitions from DT, skilled resource requirements) and external environmental conditions (e.g., external exigencies, regulatory compliance, competitor's initiatives for DT), and were negotiated with other actors (e.g., with TMT, resource managers, other executives, external technology partners). This was also relevant in terms of relative importance of the CIO is difference organizations. Finally, As Feng et al. (2021) discuss that the TMT interactions are impacted by structural power of CIO over time, the structuralist perspective was observed in HealthCo, where in the TMT influenced the CIOs decisions. However, this perspective was not very evident across other case organizations. This may be because there were no CDO position in any of our case organizations and there was no immediate position to pose conflict with CIOs contribution to DT. Nevertheless, among CIO and other actors there were no specific power conflicts were observed.

6 Implications, limitations, and future research directions

Contrary to developed economies, less is known about how IT is used in businesses in developing nations (Nissen et al. 2018). Prior studies and data collection on the CIO role also have primarily been conducted in western societies (Al-Taie et al. 2014; Hütter and Riedl 2017). This study provides a contribution by highlighting the key roles performed by CIOs from a non-Western market. Moreover, this study connects the literature on CIO roles with the roles performed in the context of DT. The study enriches DT literature by consolidating and empirically examining the role of CIOs in DT. It culls out some important and influential contributions of CIOs in DT such as, building partner alliance, technology consolidation, IT security & risk management which are not yet been discussed in prevailing literature. Further, whilst most studies derive the CIO roles from literature reviews, this study presents a distinct contribution by providing empirical evidence of the CIOs contribution to DT based on in-depth case studies. Acting as a ready reckoner, the study delineates various factors that CIOs contribute towards DT initiatives at the organizations. More specifically, the study empirically demonstrates the *how* part of DT with a diversified mix of industries. The present-day business dynamics is persistently influencing the CIO's function (Chun and Mooney 2009; Peppard et al. 2011; Bongiorno et al. 2018; Gong et al. 2019). Consequently, the role is transforming itself continuously, with the digital component gaining greater importance in overall responsibilities (Capitani 2018; Noonpakdee et al 2020; Chawla and Goyal 2021). The study

provides insights on the multidimensional task demands associated with CIOs structural position in context of DT.

This research also has some practical implications as well. While earlier studies (Menz and Scheef 2014; Whittington et al. 2019) have primarily focused on the more general strategists like Chief strategy officers (CSOs), our analysis highlights CIO's as new strategist, focusing on digital strategy. The study empirically outlines the practical issues that CIOs must address to improve the likelihood of success, optimize spending, and improve the chances of DT success. Organizations must adapt to a variety of elements in order to succeed in the digital age, including shifting consumer needs, product and service customization, operational improvements, and supply chain optimization. All these factors necessitate sophisticated leadership interventions. The research offers insights into significant areas that are and will be critical to the success of CIOs in DT initiatives. CIOs may develop and successfully coach others to respond to challenges and make DT a win at their workplace. The aspirants, who are planning their transition to CIO role may focus their attention to building these core skills to accord success in their endeavors. Further, The TMT and business leaders can use these insights to formulate strategy to achieve business objectives through DT. Knowing what to expect, they can leverage this understanding for selection of suitable candidate for the CIO position at their organization. The C-suite leaders at various levels can use the insights from the study to strategically plan the responsibilities and accountabilities for the digital projects beforehand rather than just deploying digital technologies, thereby improving the success rates. In conclusion, this study reaffirms that CIO leadership is more critical in the digital era than ever.

However, it is essential to note the limitations of the study. This study presents the inference based on semi-structured interviews, which is circumstantial. Hence, there is a possibility that some insights remain underappreciated. Though we have triangulated the results with functional participants and available information in public domain, there is still a possibility of participant and hindsight bias. Further, our understanding is from the selected cases; there could be different insights emerging from different industry types. Future researchers may take a broader sample size covering a wide diversity of industries for a more comprehensive understanding and generalization of results. Large organizations were selected for this investigation. Similar analysis on small and medium-sized enterprises (SMEs) can be performed to assess how the key role of CIOs vary in these industry types.

The strategic positioning of CIOs is influenced by their technical and business acumen (Banker et al. 2022). This study did not cover much on this aspect. The future research can analyze how the CIOs skills influenced the decision making on digital initiatives such as technology selection orientation and addressing business demands.

It is enticing to note about how CIOs prepare to embark on their DT journey. More empirical investigations are advised because DT is a phenomenon that is practice oriented. To assess how the CIO roles themselves evolve over time, longitudinal studies can be conducted to analyze novel aspects such as CIOs transitioning to newer business roles, CIOs influence on ecosystem partners, their contributions on IT security and cloud infrastructure, impact of industry-specific skills of CIOs on

the success of DT and CIO role in adoption of new technologies. Moreover, emerging global digital infrastructures are open, adaptable, and available to everyone, not limited to the specific organizations (Tilson et al. 2010; Hanelt et al. 2021). Future research might therefore examine the strategic relationship factors that CIOs take into account when developing shared digital architectures with partners. In addition, the interrelationships between the roles are not assessed, future research can examine these associations.

Finally, as country, cultural factors and legal regulations can influence the conduct and development of the CIO role (Firk et al. 2021), we suggest more studies based in emerging economies. Even if there are certain commonalities between the organizations, each one operates differently on the ground. According to Davision et al. (2023), the variations result from the many unique circumstances that surround each organization. It is probable that other organizations won't go through the same specific set of events as the case organizations. However, the variety of the case organizations selected for this study does opens the door to possible comparisons with other organizations that share a similar character, which the researchers can then further expand.

Appendix 1

Case ID	First order concepts	Second order themes	Aggregate dimensions
RC1	….for our kind of business freight is the key aspect, we started with a technical Digital framework where we reinvented the way we are doing contracting	Business Need evaluator & Business Process Innovator	Technology & Innovation
RC1	….enable all our sales team, do virtual discussions rather going physical, especially in the COVID time	Business Process Reengineer	Technology & Innovation
RC1	….so every year we are making a roadmap for two to three year sand improvise that with business	Business Strategy complementor	Strategic Management
RC1	…..All transporters were educated about this application and its knowhow	Digital Culturist	People Management
RC1	We have on boarded data science platform this year	Digital Infra & Platform enabler	IT Governance & Operations Management
RC1	We have started and the strategy we adopted that we kept on training internal team, train the trainer	Digital Talent Manager	People Management

Case ID	First order concepts	Second order themes	Aggregate dimensions
RC1	We came up with a digital framework, now sales team is able to do 10 to 12 meetings a day rather than going to site	IT & Business Liaison	Business & Functional Integration
RC1	In some cases even if it's not understood completely by them (TMT) still they have provide go ahead based on my recommendation	IT representor, Trust & Perception Builder	Communication
RC1	All fleet operator were required to be on board and convinced that it was in their favour	Negotiator	People Management
RC1	…. fitment of GPS on every vehicle was the business case..analysed..Piloted.. rolled out	Technology consolidator & Application advisor	Technology & Innovation
RC1	…	…	…
RC2	I got my self involved and kept a closed control till POC and then delegated	Agile Execution & Delivery	IT Governance & Operations Management
RC2	Drone is a technology that is about to come, the company may adopt it as a pioneer to have some advantage	Business Need evaluator & Business Process Innovator	Technology & Innovation
RC2	We are having 10, 12 companies…in sync digital needs to have a consolidated business strategy	Business Strategy complementor	Strategic Management
RC2	We tried inducing digital into DNA the organization (the group)	Digital Culturist	People Management
RC2	We need to have some people to build team internally	Digital Talent Manager	People Management
RC2	It is both IT and business team which articulates and decides what is required	IT & Business Liaison	Business & Functional Integration
RC2	The success of the first phase actually helped business having trust, now if the new proposal is coming, probably it will get through	IT representor, Trust & Perception Builder	Communication
RC2	We got the compliance done by security partner for every initiative before we go live	IT security Enabler & compliance Conformer	Conformance
RC2	There is a lot of stress on the capital investment, I have to ensure Digital projects have to be approved by chairman	Negotiator	People Management

Case ID	First order concepts	Second order themes	Aggregate dimensions
RC2	We are doing a complete supply chain integration on cost management starting from the supplier to material cost from tier two suppliers	Partner Alliancer	People Management
RC2	Deciding which product to deploy, what is the best in the market to benchmark and which solution would be better for the organization	Technology consolidator & Application advisor	Technology & Innovation
RC2	Because we are bringing a change by exploring new way of doing things, as CIO I have to convince the board about this	TMT influencer and advisor	People Management
RC2	…	…	…
RC3	We are working on four to five assignments in parallel say for outbound operations, inbound operations or data science platform with designated timelines for implementation	Agile Execution & Delivery	IT Governance & Operations Management
RC3	Most of the DT was focused on addressing improvement of internal processes	Business Process Reengineer	Technology & Innovation
RC3	….after 20 year, our business is not going to happen the way it's happening now, it will be radically transformed, we have to transform it with DT	Business Strategy complementor	Strategic Management
RC3	We failed and filled the gap. Failed Fast, improvised and aligned for delivery	Digital Culturist	People Management
RC3	It was must to have commitment from our board so that the message is communicated clearly	TMT influencer and advisor	People Management
RC3	…	…	…
RC4	As CIO, I had a critical role to make it happen faster and successful	Agile Execution & Delivery	IT Governance & Operations Management
RC4	I went to MD and discussed new patient management application	Business Need evaluator & Business Process Innovator	Technology & Innovation
RC4	With treasury automation the third party transactions are completely automated	Business Process Reengineer	Technology & Innovation

Case ID	First order concepts	Second order themes	Aggregate dimensions
RC4	One of my primary task is to device digital strategy where digital is trying to support the business strategy	Business Strategy complementor	Strategic Management
RC4	We have a strategy where we designed organisational communications top down. Every mandate, we were pushing from top	Communicator	Communication
RC4	Voice processing, image processing, you named it and we do have platform to execute it	Digital Infra & Platform enabler	IT Governance & Operations Management
RC4	…..It encompasses compensation benefits of learning and development	Digital Talent Manager	People Management
RC4	…..designed that all the applications are secured and fully compliant to government norms	IT security Enabler & compliance Conformer	Conformance
RC4	…..enabling this technology and for bringing changed processes, our prevailing application was required to be integrated to cloud	Legacy Manager & Technology Integrator	Business & Functional Integration
RC4	……Another person is there (CEO), who sponsors these new ideas for him	Negotiator	People Management
RC4	….behind that there is a team who keep on guiding us to make it happen that engagement we have created	Partner Alliancer	People Management
RC4	…	…	…
RC5	Then finding the solution, and then trying to replicate it, So bandwidth concerns would be there…. Delegate effectively	Agile Execution & Delivery	IT Governance & Operations Management
RC5	At back of mind, a strategy is there….. What is our overall picture….What is our complete basket	Business Strategy complementor	Strategic Management
RC5	We have formed a team of evangelists who act as a change agent and change manager	Digital Culturist	People Management
RC5	Infrastructure advancements and cloud based model adoptions were our priority	Digital Infra & Platform enabler	IT Governance & Operations Management

Case ID	First order concepts	Second order themes	Aggregate dimensions
RC5	I manage a team of around 40 45 people, across teams	Digital Talent Manager	People Management
RC5	…we have a sign off process actually after the functions provided their inputs to ensure all have common understanding	IT & Business Liaison	Business & Functional Integration
RC5	We started many projects parallelly including cyber security programs to make cyber security also part of digital initiatives	IT security Enabler & compliance Conformer	Conformance
RC5	Analytics is a big portfolio of dashboards designed on tablew platforms throughout	Technology consolidator & Application advisor	Technology & Innovation
RC5	…now CIO not only need to execute it, also to educate and convince that the way he's thinking is going to work	TMT influencer and advisor	People Management
RC5	…	…	…
RC6	I have to drive it faster, addressing concerns of people and company, that's very important	Agile Execution & Delivery	IT Governance & Operations Management
RC6	If the company lags the industry bench mark CIO's role becomes very critical	Business Need evaluator & Business Process Innovator	Technology & Innovation
RC6	We target to complement business for enhanced revenues primarily supported by digital technologies	Business Strategy complementor	Strategic Management
RC6	We communicated the benefits, By some initiatives man power saving is also there	Communicator	Communication
RC6	Because it's a matter of adoption of technology	Digital Culturist	People Management
RC6	…..we deliberated on how we are going to build capability on analytics	Digital Talent Manager	People Management
RC6	We're taking on audits, compliance policies, governance, certifications	IT security Enabler & compliance Conformer	Conformance
RC6	IT investment on digital has gone almost double	Negotiator	People Management
RC6	….so we have on boarded a data science company. And we have taken their platform	Partner Alliancer	People Management

Case ID	First order concepts	Second order themes	Aggregate dimensions
RC6	….descriptive, predictive, artificial intelligence, machine learning etc., but how to build was a consideration for us	Technology consolidator & Application advisor	Technology & Innovation
RC6	I have suggested many transformations which I can say got accepted by them	TMT influencer and advisor	People Management
RC6	…	…	…

Appendix 2

Research agenda (participant's interview)

Research introduction to participant (Pre Interview)

- Introduction of our research and its objectives (DT drivers, Role of CIO and implementation of best practices)
- Declaration of conformity and confidentiality

Discussion points

Organisational context

- What are your views on your organisation's digital journey so far?
- What were some of the key factors (driving forces) to take up digital initiatives?
- How do you find digital technologies impacting the overall industry? What are some of the key impact areas specific to your organization?

Role of CIO in digital transformation

- What is (was) the general focus of DT in the organization?
- Who is primarily responsible for leading digital initiatives? How is the business and IT role planned and How are the initiatives governed?
- What does the overall digital strategy of your organisation look like, is it impacting business strategy. How the strategy was crafted?
- What are some of the key initiatives taken for DT and how has the journey been so far?
- What was the top management team's (TMT Executive's) role in the DT initiatives? Their awareness about DT and how have the events progressed in this regard (In context of strategic role of executives)?
- What was (is) the overall role of Chief Information officer/Chief Digital officer in the digital Transformation? What aspects were core for DT?

- How did the lower level managers and IT team contributed to DT initiatives?
- What were some of the challenges faced in DT initiatives and how were they addressed?

How to successfully implement DT

- What was the implementation methodology followed for implementing DT, was it done through your own team or partner team, role of consultants?
- What are some of the major contributions that DT initiatives have brought so far (such as business processes improvements/revenue/culture/ways of working etc.)?
- What aspects do you see that governed success or a failure to the DT initiatives? What are some of the key points that you would like to recommend for effective implementation of DT?
- How was the knowledge built on DT, how the cultural adoption was done, how was IT security aspect managed. What were some of the challenges faced?
- What would you differently if you have to do it again?

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